Priyanshu Ranjan

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Professional Summary

Computer Science undergraduate specializing in AI & Machine Learning with a CGPA of **8.76/10**. Built AI-enabled applications and ML pipelines spanning LLM integration, customer churn prediction, computer vision, NLP, and model-serving APIs. Experienced in data preprocessing, feature selection, model evaluation, explainability, and integrating AI services into deployed applications.

Education

Vellore Institute of Technology **Sep 2023 – Expected 2027**
B.Tech in Computer Science & Engineering (AI & ML Specialization) | ***CGPA: 8.76/10***

Projects

AI HealthBot - Symptom Assessment Platform **Jan – Mar 2026**

LLM-Powered Healthcare Support — React, TypeScript, Node.js, Supabase, Gemini 2.5 Pro

- Built a bilingual English/Hindi healthcare-support application using **Gemini 2.5 Pro**, a **9-step symptom assessment** flow, contextual follow-up questions, and **3-level risk triage** (Low, Moderate, High).
- Designed structured Gemini prompts with standardized symptom-analysis sections, emergency-care guidance for high-risk symptoms, and follow-up questions when symptom descriptions are too vague to assess.
- Integrated AI recommendations through **Supabase Edge Functions** and PostgreSQL-backed storage for health-history and application workflows; positioned the application as preliminary decision support rather than medical diagnosis.

Customer Churn Prediction Platform **Aug – Sept 2026**

Supervised ML & Explainable AI — Python, Pandas, Scikit-learn, XGBoost, SHAP, FastAPI

- Prepared the **7,043-customer Telco Churn dataset** using preprocessing, feature selection, and a stratified train-test split; excluded identifier, geographic, and outcome-related fields from model inputs.
- Benchmarked **5 classifiers** and tuned XGBoost with **5-fold stratified cross-validation**; the repository reports **84.81% ROC-AUC** and **60.06% F1** on its test split.
- Integrated **SHAP** explanations and risk-based retention recommendations into a FastAPI prediction service, with a Streamlit interface and Tableau analytics dashboard.

AI Smart Retail - Customer Intelligence Platform **Jun – Jul 2026**

Multi-Model AI Application — MobileNetV2, OpenCV, TF-IDF, FastAPI, React

- Built a computer-vision pipeline combining **MobileNetV2** product classification across **5 retail categories** with OpenCV/Haar-Cascade face detection and image-derived feature matching for customer lookup.
- Implemented TF-IDF/Logistic Regression sentiment analysis and a hybrid FAQ chatbot combining rule-based matching, an ML intent classifier, and **20+ intents.**
- Exposed computer-vision, sentiment, and chatbot services through API-key-protected **FastAPI endpoints** with SQLite/ORM persistence; integrated a React dashboard, Docker Compose files, pytest modules, and GitHub Actions

Technical Skills

Languages	Python, SQL, Java
Machine Learning	Scikit-learn, XGBoost, PyTorch, Model Evaluation
AI / NLP / CV	SHAP, TF-IDF, OpenCV
Data Processing	Pandas, NumPy, EDA, Feature Selection
AI Applications	Flask, FastAPI, REST APIs, Supabase Edge Functions
Visualization / Tools	Power BI, Tableau, React, TypeScript, Git, Google Colab

Certifications & Achievements

- **Oracle Cloud Infrastructure 2025 Certified Data Science Professional** — Oracle
- **Oracle Agentic AI Foundations Associate (1Z0-1157-26)** — Oracle
- **Applied Machine Learning in Python** — Certificate
- **TCS CodeVita Season 13 – Global Rank: 1206** — Proof
- **Graph Camp – AlgoUniversity** — Certificate